Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

APPLIED BUSINESS M2

Practical – VII

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Practical 7: Sampling and Sampling Distributions

Part A: Survey and Sampling

1. Select a real-life business topic for conducting a survey.

Topic: Student Exam Performance and Test Preparation Survey

Purpose: Students ke exam scores aur test preparation ke beech data ko analyze karna.

2. Prepare a questionnaire containing at least 5 questions related to the selected topic.

What is your gender?
What is your parental level of education?
What type of lunch do you usually have?
Did you complete a test preparation course?
What is your Mathematics score?

3. Collect responses from at least 50 people.

4. Store the collected data in a CSV file or Pandas DataFrame.

INPUT:-

```
import pandas as pd
# Load the dataset
df = pd.read_csv("StudentsPerformance.csv")
# Display dataset information
print("Number of records:", len(df))
print("Number of columns:", len(df.columns))
# Store the data in a Pandas DataFrame
print("\nDataset stored successfully in DataFrame.")
```

OUTPUT:-

```
Number of records: 1000
Number of columns: 8

Dataset stored successfully in DataFrame.
```

5. Display the first five records of the collected data.

INPUT:-

```
print(df.head())
```

OUTPUT:-

```
   gender race/ethnicity parental level of education         lunch  \
0  female        group B           bachelor's degree      standard
1  female        group C                some college      standard
2  female        group B             master's degree      standard
3    male        group A          associate's degree  free/reduced
4    male        group C                some college      standard

  test preparation course  math score  reading score  writing score
0                    none          72             72             74
1               completed          69             90             88
2                    none          90             95             93
3                    none          47             57             44
4                    none          76             78             75
```

6. Identify the population and sample for the survey.

INPUT:-

```
population_size = len(df)
sample = df.sample(n=50, random_state=42)
print("Population size:", population_size)
print("Sample size:", len(sample))
```

OUTPUT:-

```
Population size: 1000
Sample size: 50
```

7. Identify the sampling technique used to select the sample.

INPUT:-

```
sample = df.sample(n=50, random_state=42)
print("Sampling Technique: Simple Random Sampling")
print("Number of selected students:", len(sample))
```

OUTPUT:-

```
Sampling Technique: Simple Random Sampling
Number of selected students: 50
```

8. Identify the numerical and categorical variables in the collected data.

INPUT:-

```
numerical_variables = df.select_dtypes(include="number").columns
categorical_variables = df.select_dtypes(include="object").columns
print("Numerical Variables:")
print(list(numerical_variables))
print("\nCategorical Variables:")
print(list(categorical_variables))
```

OUTPUT:-

```
Numerical Variables:
['math score', 'reading score', 'writing score']

Categorical Variables:
['gender', 'race/ethnicity', 'parental level of education', 'lunch', 'test preparation course']
```

9. Select one numerical variable for further analysis.

INPUT:-

```
selected_variable = sample["math score"]
print("Selected numerical variable: Math Score")
```

OUTPUT:-

```
Selected numerical variable: Math Score
```

10. Calculate the sample size, mean, median, and standard deviation of the selected Variable.

INPUT:-

```
sample_size = selected_variable.size
mean = selected_variable.mean()
median = selected_variable.median()
std_dev = selected_variable.std()
print("Sample Size:", sample_size)
print("Mean:", round(mean, 2))
print("Median:", round(median, 2))
print("Standard Deviation:", round(std_dev, 2))
```

OUTPUT:-

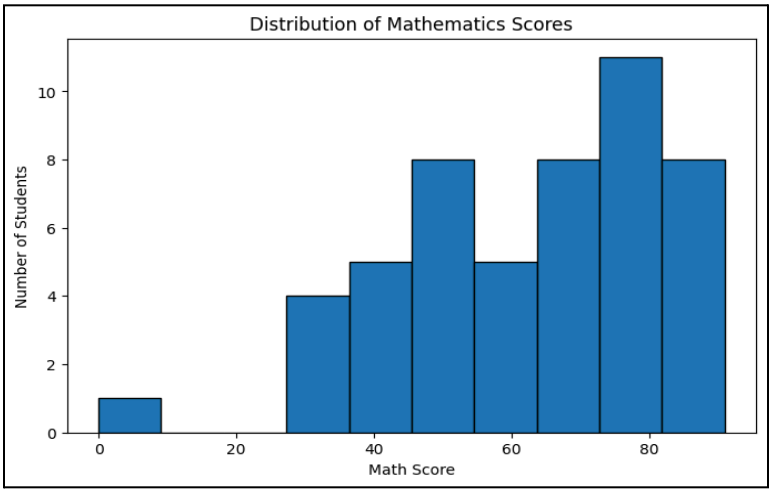
```
Sample Size: 50
Mean: 62.68
Median: 66.0
Standard Deviation: 19.13
```

11. Create a suitable chart to display the collected data.

INPUT:-

```
import matplotlib.pyplot as plt
plt.figure(figsize=(8, 5))
plt.hist(selected_variable, bins=10, edgecolor="black")
plt.title("Distribution of Mathematics Scores")
plt.xlabel("Math Score")
plt.ylabel("Number of Students")
plt.show()
```

OUTPUT:-



12. Write two observations from the survey data.

Observation 1: The Mathematics scores are distributed across different score ranges, showing variation in student performance.

Observation 2: The mean and median of Mathematics scores provide an indication of the typical performance of the selected students.

Part B: Sampling Distribution and Central Limit Theorem

13. Select a numerical variable from the collected data.

INPUT:-

```
# Select Math Score as the numerical variable
math_scores = df["math score"]
print("Selected numerical variable: Math Score")
```

OUTPUT:-

```
Selected numerical variable: Math Score
```

14. Select a sample of 30 observations randomly from the data.

INPUT:-

```
# Select 30 observations randomly
sample_30 = math_scores.sample(n=30, random_state=42)

print("Random sample of 30 Math Scores:")
print(sample_30.to_list())
```

OUTPUT:-

```
Random sample of 30 Math Scores:
[91, 53, 80, 74, 84, 81, 69, 54, 87, 51, 45, 30, 67, 49, 85, 65, 53, 55, 48, 56, 79, 43, 45, 36, 80, 80, 68, 40, 34, 49]
```

15. Calculate the mean of the selected sample.

INPUT:-

```
sample_mean_30 = sample_30.mean()
print("Mean of the sample of 30 observations:", round(sample_mean_30, 2))
```

OUTPUT:-

```
Mean of the sample of 30 observations: 61.03
```

16. Repeat the random sampling process 100 times.

17. Calculate and store the mean of each sample.

INPUT:-

```
import numpy as np
sample_means = []
for i in range(100):
    random_sample = math_scores.sample(n=30)
    mean_value = random_sample.mean()
    sample_means.append(mean_value)
print("Number of sample means:", len(sample_means))
```

OUTPUT:-

```
Number of sample means: 100
```

18. Display the first 10 sample means.

INPUT:-

```
print("First 10 Sample Means:")
for i, mean_value in enumerate(sample_means[:10], start=1):
   print(f"Sample {i}: {mean_value:.2f}")
```

OUTPUT:-

```
First 10 Sample Means:
Sample 1: 65.30
Sample 2: 69.33
Sample 3: 66.03
Sample 4: 68.90
Sample 5: 61.53
Sample 6: 69.20
Sample 7: 66.90
Sample 8: 66.33
Sample 9: 75.63
Sample 10: 66.17
```

19. Calculate the mean of all the sample means.

INPUT:-

```
mean_of_sample_means = np.mean(sample_means)
print("Mean of all sample means:", round(mean_of_sample_means, 2))
```

OUTPUT:-

```
Mean of all sample means: 66.29
```

20. Calculate the standard deviation of the sample means.

INPUT:-

```
std_sample_means = np.std(sample_means, ddof=1)
print("Standard Deviation of Sample Means:", round(std_sample_means, 2))
```

OUTPUT:-

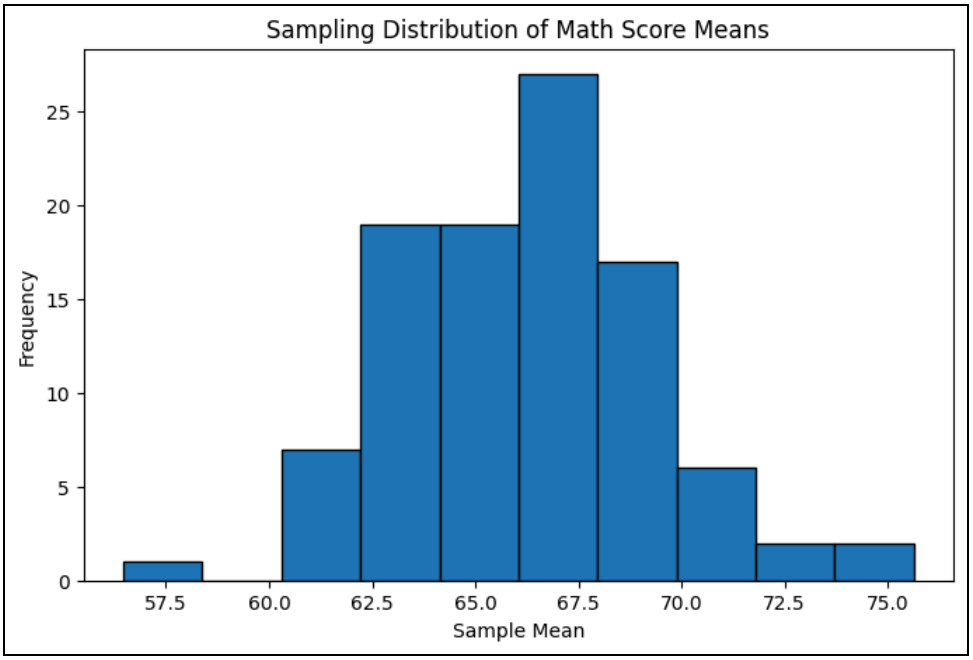
```
Standard Deviation of Sample Means: 3.1
```

21. Create a histogram of the sample means.

INPUT:-

```
import matplotlib.pyplot as plt
plt.figure(figsize=(8, 5))
plt.hist(sample_means, bins=10, edgecolor="black")
plt.title("Sampling Distribution of Math Score Means")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.show()
```

OUTPUT:-



22. Observe the shape of the sampling distribution.

Observation:

The sampling distribution of the sample means is approximately bell-shaped and concentrated around the mean.

23. Compare the original sample mean with the mean of the sampling distribution.

INPUT:-

```
print("Original Sample Mean:", round(sample_mean_30, 2))
print("Mean of Sampling Distribution:", round(mean_of_sample_means, 2))
```

OUTPUT:-

```
Original Sample Mean: 61.03
Mean of Sampling Distribution: 66.29
```

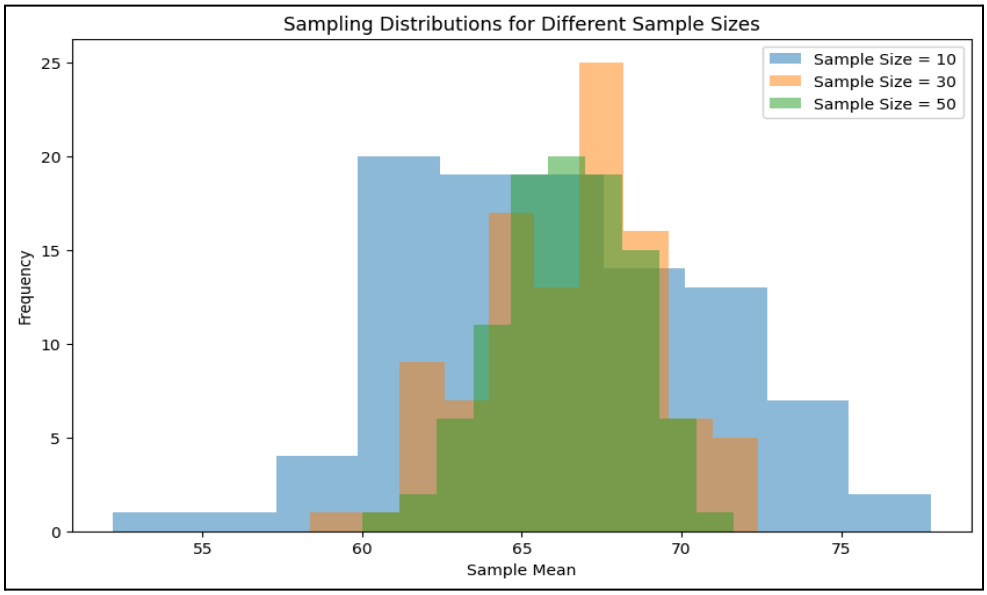
Explanation:-The original sample mean and the mean of the sampling distribution are expected to be relatively close. This shows that repeated sample means tend to center around the population mean.

24. Compare the sampling distributions obtained for different sample sizes.

INPUT:-

```
sample_sizes = [10, 30, 50]
plt.figure(figsize=(10, 6))
for size in sample_sizes:
    means = []
    for i in range(100):
        random_sample = math_scores.sample(n=size)
        means.append(random_sample.mean())
    plt.hist(means, bins=10, alpha=0.5, label=f"Sample Size = {size}")
plt.title("Sampling Distributions for Different Sample Sizes")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```

OUTPUT:-



Observation: As the sample size increases, the sampling distribution becomes narrower and the sample means become more concentrated around the population mean.

25. Calculate the standard error of the sample mean.

INPUT:-

```
population_std = math_scores.std()
standard_error = population_std / np.sqrt(30)
print("Population Standard Deviation:", round(population_std, 2))
print("Standard Error of Sample Mean:", round(standard_error, 2))
```

OUTPUT:-

```
Population Standard Deviation: 15.16
Standard Error of Sample Mean: 2.77
```

26. Explain how the Central Limit Theorem is demonstrated through the experiment.

Central Limit Theorem (CLT):

The Central Limit Theorem states that when random samples are repeatedly taken from a population, the distribution of their sample means tends to become approximately normal as the sample size increases.

In this experiment, 30 observations were randomly selected and the sampling process was repeated 100 times. The means of these samples formed a sampling distribution that was approximately bell-shaped. This demonstrates the Central Limit Theorem.

27. Use the sample mean to estimate the population mean.

INPUT:-

```
population_mean = math_scores.mean()
print("Population Mean:", round(population_mean, 2))
print("Sample Mean:", round(sample_mean_30, 2))
print("Estimated Population Mean:", round(sample_mean_30, 2))
```

OUTPUT:-

```
Population Mean: 66.09
Sample Mean: 61.03
Estimated Population Mean: 61.03
```

Explanation: The mean of the selected sample is used as an estimate of the population mean.

28. Write a conclusion based on the sampling distribution

Conclusion: The sampling distribution of the Math Scores was obtained by repeatedly selecting random samples of 30 observations. The distribution of sample means was approximately bell-shaped and centered around the population mean. As the sample size increased, the sample means became less variable. The experiment demonstrates the Central Limit Theorem and shows that a sample mean can be used to estimate the population mean.